Synoptic Question 16: The Physics of a Fission Power Plant

Reading Passage

Read the following article and then answer the questions that follow.

Nuclear fission power plants generate electricity by harnessing the energy released when the nucleus of a heavy atom, such as Uranium-235 (^{235}U), is split. This process is initiated when the nucleus in a fuel rod absorbs a slow-moving (thermal) neutron, causing it to become unstable and break apart into two smaller "daughter" nuclei and several new fast-moving neutrons.

This splitting, or fission, releases a tremendous amount of energy. This energy is a result of a "mass defect", where the total mass of the products is less than the mass of the original ^{235}U nucleus and the incident neutron. The missing mass is converted into energy, primarily the kinetic energy of the daughter nuclei and neutrons, according to Einstein's equation $E = mc^2$.

The fission process also releases intense, highly penetrating gamma radiation and radioactive particles. To ensure safety, the entire reactor core is enclosed in a massive containment structure made of thick concrete and steel. This shielding is designed to absorb the radiation, preventing it from escaping into the environment. The effectiveness of a shield is determined by its ability to attenuate, or reduce the intensity of, the radiation passing through it.

In a Pressurised Water Reactor (PWR), the thermal energy from fission is transferred to a primary water coolant loop. This high-pressure water is then passed through a heat exchanger (steam generator), where it boils a separate, secondary loop of water to create high-pressure steam. This steam spins a turbine, which in turn drives a generator to produce electricity.

Data:

Data for a typical PWR and a specific fission reaction:

Table 1: Data for Fission Analysis

Parameter	**Value**
Mass of Uranium-235 nucleus	3.9030×10^{-25} kg
Mass of Krypton-92 nucleus	1.5259×10^{-25} kg
Mass of Barium-141 nucleus	2.3389×10^{-25} kg
Mass of a neutron, m_n	1.6749×10^{-27} kg
Average energy released per ^{235}U fission, E_{avg}	3.20×10^{-11} J
Total thermal power output of reactor, P_{th}	3400 MW
Total electrical power output of plant, P_{elec}	1100 MW
Mass of ^{235}U in one fuel rod	1.50 kg
Half-life of ^{235}U, $T_{1/2}$	7.04×10^8 years
Linear attenuation coefficient of shielding, μ_{shield}	0.45 m^{-1}
Specific heat capacity of water, c_w	4200 J kg^{-1} K^{-1}
Mass flow rate of primary coolant, m_{flow}	15000 kg s^{-1}
Speed of light, c	3.00×10^8 m s^{-1}
Avogadro's constant, N_A	6.02×10^{23} mol^{-1}
Molar mass of ^{235}U	0.235 kg mol^{-1}

Questions

(a) One of the many possible fission reactions is:

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{92}_{36}Kr + ^{141}_{56}Ba + \text{neutrons}$$

(i) Determine the number of neutrons released in this specific reaction.

(ii) Calculate the mass defect (Δm) for this reaction.

(iii) Calculate the energy released (in joules) from this single reaction.

(b) The reactor core produces a total thermal power of 3400 MW.

(i) Using the average energy released per fission, E_{avg}, calculate the total number of fission reactions occurring in the reactor core every second.

(ii) Calculate the total mass of ^{235}U consumed in one day.

(c) Consider a single 1.50 kg fuel rod composed entirely of ^{235}U.

(i) Calculate the number of ^{235}U atoms in the fuel rod.

(ii) Calculate the decay constant, λ, for ^{235}U in s^{-1}.

(iii) Calculate the initial activity (in Bq) of this fuel rod due to the decay of ^{235}U.

(d) Explain why the reactor core must be heavily shielded with materials like concrete and lead.

(e) The shielding's effectiveness is measured by its attenuation of gamma rays. The intensity I of radiation after passing through a thickness x of a material is given by the exponential decay equation $I = I_0 e^{-\mu x}$, where μ is the linear attenuation coefficient.

(i) Show that the half-value thickness (HVT, $x_{1/2}$), the thickness required to halve the intensity, is related to μ by $x_{1/2} = \frac{\ln(2)}{\mu}$.

(ii) Using the data for the reactor shielding, calculate the thickness of shielding required to reduce the gamma ray intensity to 0.010% of its initial value.

(f) The thermal energy is transferred to the primary water coolant.

(i) Using the formula $P = mc\Delta T$, where m is the mass per second (mass flow rate), calculate the temperature increase, ΔT, of the primary coolant as it passes through the reactor core.

(ii) Calculate the overall efficiency of the power plant in converting thermal energy into electrical energy.

(g) The steam turbine spins a generator. Briefly describe the principle of electromagnetic induction by which the generator produces an alternating e.m.f.

Solutions

(a) (i) Let x be the number of neutrons.
Mass number: $235 + 1 = 92 + 141 + x(1) \implies 236 = 233 + x \implies x = 3$.
3 neutrons are released.

(ii) Mass of reactants $= m_U + m_n = (3.9030 \times 10^{-25}) + (1.6749 \times 10^{-27}) = 3.9197 \times 10^{-25}$ kg.
Mass of products $= m_{Kr} + m_{Ba} + 3m_n = (1.5259 \times 10^{-25}) + (2.3389 \times 10^{-25}) + 3(1.6749 \times 10^{-27})$
Mass of products $= 3.8648 \times 10^{-25} + 0.0502 \times 10^{-25} = 3.9150 \times 10^{-25}$ kg.
$\Delta m = \text{Mass}_{reactants} - \text{Mass}_{products} = (3.9197 - 3.9150) \times 10^{-25} = 4.7 \times 10^{-27}$ kg.

(iii) $E = \Delta mc^2 = (4.7 \times 10^{-27}) \times (3.00 \times 10^8)^2 = 4.23 \times 10^{-10}$ J.

(b) (i) $P_{th} = 3400$ MW $= 3.40 \times 10^9$ W (J s^{-1}).
Reactions per second $= \frac{P_{th}}{E_{avg}} = \frac{3.40 \times 10^9}{3.20 \times 10^{-11}} = 1.06 \times 10^{20}$ s^{-1}.

(ii) Fissions in one day $= (1.06 \times 10^{20}) \times (24 \times 3600) = 9.16 \times 10^{24}$ fissions.
Mass of one ^{235}U atom $\approx 3.9030 \times 10^{-25}$ kg.
Mass consumed = (Atoms consumed) $\times$ (Mass per atom)
Mass consumed $= (9.16 \times 10^{24}) \times (3.9030 \times 10^{-25}) = 3.575$ kg.

(c) (i) Number of moles $n = \frac{\text{mass}}{\text{molar mass}} = \frac{1.50}{0.235} = 6.38$ mol.
Number of atoms $N = n \times N_A = (6.38) \times (6.02 \times 10^{23}) = 3.84 \times 10^{24}$ atoms.

(ii) $T_{1/2} = (7.04 \times 10^8 \text{ years}) \times (365.25 \times 24 \times 3600) = 2.22 \times 10^{16}$ s.
$\lambda = \frac{\ln(2)}{T_{1/2}} = \frac{0.693}{2.22 \times 10^{16}} = 3.12 \times 10^{-17}$ s^{-1}.

(iii) Activity $A = \lambda N = (3.12 \times 10^{-17}) \times (3.84 \times 10^{24}) = 1.20 \times 10^8$ Bq.

(d) The fission process releases intense, highly penetrating radiation (neutrons and gamma rays) which is extremely harmful to living organisms. The shielding is required to absorb this radiation and reduce the intensity to safe levels outside the reactor, protecting the workers and the environment.

(e) (i) For HVT, $x = x_{1/2}$ and the intensity $I = \frac{I_0}{2}$.
$\frac{I_0}{2} = I_0 e^{-\mu x_{1/2}}$
$\frac{1}{2} = e^{-\mu x_{1/2}}$
Take natural log (ln) of both sides: $\ln(\frac{1}{2}) = -\mu x_{1/2}$
$-\ln(2) = -\mu x_{1/2} \implies x_{1/2} = \frac{\ln(2)}{\mu}$. (Proven)

(ii) We want the final intensity I to be 0.010% of I_0.
$I = 0.00010 \times I_0$.
$0.00010 \times I_0 = I_0 e^{-\mu x}$
$0.00010 = e^{-0.45x}$
$\ln(0.00010) = -0.45x$
$-9.21 = -0.45x$
$x = \frac{9.21}{0.45} = 20.47$ m ≈ 20.5 m.

(f) (i) $P = (m_{flow})c_w \Delta T \implies \Delta T = \frac{P}{m_{flow} c_w}$
$\Delta T = \frac{3.40 \times 10^9}{(15000)(4200)} = 53.97$ K (or 54.0 °C).

(ii) Efficiency $\eta = \frac{\text{Useful Power Out}}{\text{Total Power In}} = \frac{P_{elec}}{P_{th}}$
$\eta = \frac{1100 \text{ MW}}{3400 \text{ MW}} = 0.3235 \approx 32.4\%$.

(g) The steam turbine spins a large coil of wire within a magnetic field (or spins a magnet within a set of coils). This rotation causes the magnetic flux linkage through the coils to change continuously over time. According to Faraday's Law of Induction, a changing magnetic flux induces an electromotive force (e.m.f.) in the coils. The continuous rotation naturally produces an alternating (sinusoidal) e.m.f., driving an alternating current.

[Diagram of a Pressurised Water Reactor (PWR) showing core, containment, coolant loops, and turbine]

Figure 1: Schematic of a Pressurised Water Reactor (PWR) Plant

Cambridge International AS & A Level Physics (9702) Paper 4 Structured Questions Variant 2 (Higher Difficulty)

Question 16V2: Advanced Physics of a Fission Power Station

Additional data for Variant 2:

- Thermal power of reactor, $P_{\text{th}} = 3.40 \times 10^9\,\text{W}$
- Electrical output power, $P_{\text{elec}} = 1.10 \times 10^9\,\text{W}$
- Mean energy released per fission, $E_f = 3.20 \times 10^{-11}\,\text{J}$
- Energy released per captured neutron in control rod material, $E_{\text{abs}} = 8.0 \times 10^{-13}\,\text{J}$
- Mass flow rate of primary coolant, $\dot{m} = 1.50 \times 10^4\,\text{kg}\,\text{s}^{-1}$
- Specific heat capacity of water, $c = 4200\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$
- Turbine entry steam temperature $= 275\,°\text{C} = 548\,\text{K}$
- Condenser temperature $= 35\,°\text{C} = 308\,\text{K}$
- Linear attenuation coefficient for shielding material, $\mu = 0.45\,\text{m}^{-1}$
- Speed of light, $c_0 = 3.00 \times 10^8\,\text{m}\,\text{s}^{-1}$

Questions (Total = 50 marks)

(a) For the fission reaction

$$^{235}_{92}\text{U} + ^{1}_{0}\text{n} \rightarrow ^{92}_{36}\text{Kr} + ^{141}_{56}\text{Ba} + x\,^{1}_{0}\text{n}$$

(i) Determine x. [1]

(ii) State why emitted neutrons must usually be moderated in a thermal reactor. [2]

(iii) Explain why delayed neutrons are important for safe reactor control. [2]

(b) The station operates at thermal power $P_{\text{th}} = 3.40 \times 10^9\,\text{W}$.

(i) Calculate the fission rate in the core. [2]

(ii) If each fission emits 3 neutrons and, at steady power, exactly one neutron per fission sustains the next fission, determine the number of neutrons per second that must be removed by leakage and absorbers combined. [2]

(iii) A control rod insertion increases neutron absorption by $1.0 \times 10^{19}\,\text{s}^{-1}$. Estimate the immediate thermal power reduction caused by this insertion, assuming each absorbed neutron would otherwise have caused one fission. [3]

(c) Shielding design uses

$$I = I_0 e^{-\mu x}.$$

(i) Derive an expression for thickness x in terms of transmission fraction $T = I/I_0$. [2]

(ii) Calculate x needed for $T = 1.0 \times 10^{-6}$. [2]

(iii) Open-ended: discuss, with physics reasons, why practical biological shielding is often multilayered rather than a single material, even when a single-value μ model is provided. [4]

(d) Primary coolant enters the core at 290 °C and exits at a higher temperature.

(i) Calculate the temperature rise of the primary coolant across the core. [2]

(ii) Open-ended: explain two engineering reasons why pressurised water is used in the primary loop rather than allowing it to boil in the reactor vessel in this design. [4]

(e) The plant converts thermal energy to electrical energy.

(i) Determine the actual thermal efficiency of the station. [1]

(ii) Calculate the maximum (Carnot) efficiency between 548 K and 308 K. [2]

(iii) Hence comment quantitatively on how closely the station approaches the Carnot limit. [2]

(iv) Open-ended: identify and explain two major sources of irreversibility in real turbine-generator systems that reduce efficiency below the Carnot value. [4]

(f) **Extended response (synoptic).** [15]

A proposal is made to uprate the reactor thermal power by 10% while keeping the same core geometry and the same secondary-side condenser temperature.

Discuss the physics implications for reactor kinetics, heat transfer, thermal stress, safety margins, and electrical output.

Your answer should include:

- at least one justified calculation,
- clear links between microscopic nuclear processes and macroscopic plant performance,
- at least two reasoned safety constraints that may limit achievable uprate.

Sample Solutions (Variant 2)

(a) (i) Mass number conservation: $235 + 1 = 92 + 141 + x \Rightarrow x = 3$.

(ii) Fast neutrons have low fission probability in ^{235}U thermal-spectrum designs; moderation reduces neutron kinetic energy, increasing fission cross-section and sustaining chain reaction.

(iii) Delayed neutrons (from fission-product decay) appear later than prompt neutrons, increasing effective response time of reactor power, making control by rods/feedback feasible and stable.

(b) (i) Fission rate

$$\dot{N}_f = \frac{P_{\text{th}}}{E_f} = \frac{3.40 \times 10^9}{3.20 \times 10^{-11}} = 1.06 \times 10^{20}\,\text{s}^{-1}.$$

(ii) Neutrons produced per second $= 3\dot{N}_f$. Neutrons needed to sustain chain $= \dot{N}_f$. Removed neutrons

$$\dot{N}_{\text{rem}} = 2\dot{N}_f = 2.12 \times 10^{20}\,\text{s}^{-1}.$$

(iii) Extra absorption rate $\Delta\dot{N} = 1.0 \times 10^{19}\,\text{s}^{-1}$ gives lost fissions at same rate.

$$\Delta P_{\text{th}} = \Delta\dot{N}\,E_f = (1.0 \times 10^{19})(3.20 \times 10^{-11}) = 3.2 \times 10^8\,\text{W}.$$

So immediate reduction is about 320 MW, about 9.4% of nominal thermal power.

(c) (i) From $T = e^{-\mu x}$,

$$\ln T = -\mu x \Rightarrow x = -\frac{\ln T}{\mu}.$$

(ii)

$$x = -\frac{\ln(10^{-6})}{0.45} = \frac{13.8155}{0.45} = 30.7\,\text{m}.$$

(iii) Multilayer shielding is used because different radiations interact differently: hydrogen-rich materials slow neutrons; boron/cadmium capture thermal neutrons; dense high-Z layers attenuate gamma rays; concrete adds structural, thermal and cost advantages. Secondary gamma production and activation products also motivate staged design.

(d) (i)

$$\Delta T = \frac{P_{\text{th}}}{\dot{m}c} = \frac{3.40 \times 10^9}{(1.50 \times 10^4)(4200)} = 54.0\,\text{K}.$$

(ii) Pressurisation raises boiling point well above operating temperature, preventing void formation that would alter moderation and reactivity; liquid water has high heat capacity and enables stable, high-rate heat removal with predictable flow and lower risk of local dry-out in fuel channels.

(e) (i)

$$\eta_{\text{actual}} = \frac{P_{\text{elec}}}{P_{\text{th}}} = \frac{1.10 \times 10^9}{3.40 \times 10^9} = 0.324\ (32.4\%).$$

(ii)

$$\eta_{\text{Carnot}} = 1 - \frac{T_c}{T_h} = 1 - \frac{308}{548} = 0.438\ (43.8\%).$$

(iii) Fraction of Carnot achieved

$$\frac{\eta_{\text{actual}}}{\eta_{\text{Carnot}}} = \frac{0.324}{0.438} = 0.74.$$

So the plant achieves about 74% of the ideal limit.

(iv) Irreversibility sources include turbine blade friction/turbulence (entropy production), finite-temperature heat transfer in steam generator and condenser, generator copper/core losses, pump work, moisture losses in low-pressure turbine stages, and parasitic auxiliary loads.

(f) Indicative extended response:

A 10% uprate gives new thermal power $P'_{\text{th}} = 1.10 \times 3.40 = 3.74 \times 10^9\,\text{W}$. If efficiency were unchanged, electrical output would rise to about

$$P'_{\text{elec}} \approx 0.324 \times 3.74 \times 10^9 = 1.21 \times 10^9\,\text{W}.$$

Fission rate must also rise by 10%, increasing neutron flux and fuel burnup rate. Higher flux elevates fuel temperature and cladding temperature gradients, raising thermal stress and potentially reducing material lifetime. Coolant outlet temperature or required mass flow must rise; if flow stays fixed, core ΔT rises by 10% (to about 59 K), narrowing departure-from-nucleate-boiling margin.

Reactor kinetics become more sensitive because higher power density amplifies reactivity-feedback consequences (fuel Doppler broadening, moderator density effects). Control rods and boron system must retain shutdown margin under hot conditions and transient scenarios.

Safety constraints likely include: (1) peak fuel/cladding temperature limits to avoid damage and excessive fission-product release risk; (2) coolant pressure and critical heat flux margins to avoid boiling crisis; additionally containment heat removal and emergency core cooling performance must remain compliant. Thus uprate may be feasible only with hardware upgrades, revised operating envelope, and demonstration that accident analyses still meet dose and integrity criteria.

Mark Scheme (Variant 2, CIE 9702 P4 Style)

Notation: M = method mark, A = accuracy mark, B = independent point, QWC = quality of written communication.

5 (i) $x = 3$ (B1)

(ii) Any two: moderation lowers neutron speed/energy; increases fission cross-section in thermal reactor; supports sustained chain reaction (B1+B1)

(iii) Delayed neutrons increase control timescale / permit stable feedback and rod action (B1+B1)

7 (i) $\dot{N}_f = P/E_f$ (M1), $1.06 \times 10^{20}\,\mathrm{s}^{-1}$ (A1)

(ii) Correct neutron balance setup (M1), $2.12 \times 10^{20}\,\mathrm{s}^{-1}$ (A1)

(iii) Uses $\Delta P = \Delta\dot{N}\,E_f$ (M1), obtains $3.2 \times 10^8\,\mathrm{W}$ / $320\,\mathrm{MW}$ (A1), sensible percentage comment (A1)

8 (i) Rearrangement to $x = -\ln T/\mu$ (M1A1)

(ii) Substitution with $T = 10^{-6}$ (M1), $x \approx 30.7\,\mathrm{m}$ (A1)

(iii) Open-ended points (max 4B): distinct treatment of neutrons/gamma; role of hydrogenous moderator layer; role of absorber (e.g. boron); role of dense high-Z layer; secondary gamma/activation consideration; practical structural/cost argument.

6 (i) Uses $\Delta T = P/(\dot{m}c)$ (M1), $54.0\,\mathrm{K}$ (A1)

(ii) Open-ended reasons (max 4B): suppress boiling/voids; maintain predictable moderation/reactivity; improve heat transfer stability; avoid dry-out/hot spots; operational safety/control argument.

9 (i) $\eta_{\text{actual}} = 0.324$ or 32.4% (B1)

(ii) Correct Carnot expression (M1), 0.438 or 43.8% (A1)

(iii) Compares efficiencies quantitatively, e.g. 74% of Carnot (B1+B1)

(iv) Open-ended irreversibilities (max 4B): turbine friction/turbulence; finite-ΔT heat exchange; generator electrical losses; auxiliary/pumping losses; wetness losses; entropy-generation explanation.

15 Level-based marking for structured extended response:

- **11–15 marks:** coherent, well-structured response with relevant physics throughout; includes justified calculation(s); clear micro-to-macro linkage; at least two well-explained safety constraints; minimal errors. (QWC2 + up to B13)
- **6–10 marks:** partially structured with several correct physics points; at least one valid calculation attempt; some safety discussion but limited depth or linkage. (QWC1 + up to B9)
- **1–5 marks:** fragmented but contains some correct statements about power uprate implications; little quantitative support. (up to B5)
- **0 marks:** no creditworthy content.